Late-Layer Geometric Resistance Under In-Context Knowledge Conflict

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Status: workshop-note draft · CPU/MPS experiments · 2026-07-06 **Scope:** a narrow, falsification-heavy interpretability finding — see Section 5 before citing.

Abstract

When a document in context contradicts a language model's parametric knowledge, does that conflict leave a measurable trace in the residual stream? We test this on two instruction-tuned models (Qwen3-4B-Instruct, Phi-3-mini) across three relation types (country→capital, company→founder, book→author) and find a real, replicated signal: at a late layer (~83% depth), the hidden state for a same-entity contradiction is geometrically further from a clean baseline than switching to a different entity entirely is; at a mid layer (~33% depth), there is no such gap. We then falsify our own first two explanations for *why* the effect varies in size — first that it tracks per-fact parametric memorization strength (rigidity hypothesis, refuted: Spearman rho=0.006, n=45), then that it tracks the model successfully outputting the false claim (active-lie hypothesis, refuted: resisting the contradiction costs *more* geometric disruption than complying with it, n=73 vs 62). The mid-layer null is itself reconciled with prior work: a *trained* probe recovers a real but modest signal (AUROC 0.642) where our untrained geometric check found none, consistent with — but quantitatively short of — Zhao et al. (2024)'s reported ~90% mid-layer probe accuracy.

1. Setup

Three relations, 15 facts each (45 total), each posed as `Document: {fact}. Q: {question} A:` with either the true tail (congruent) or another fact's real tail substituted (contradictory, a plausible hard negative rather than gibberish). Two models confirmed to actually follow in-context overrides before use (checked directly, not assumed): GPT-2-base and TinyLlama-1.1B-Chat both failed this premise outright and were excluded; Qwen3-4B-Instruct and Phi-3-mini-4k-instruct both pass. Llama-3 itself is gated on Hugging Face (manual approval required) and was not accessible in this environment.

2. Results

2.1 Real-LLM parametric recall (baseline sanity check)

A linear probe on GPT-2 hidden states, trained on 20 countries' true/false capital sentences, reaches AUROC 0.980 on 10 held-out countries — confirms parametric facts are linearly decodable and generalize to unseen entities, before testing anything harder.

2.2 Late-layer decoherence under in-context contradiction

Cosine similarity between congruent and contradictory hidden states, same entity, vs. cosine similarity between two different entities' congruent hidden states (the "does a lie perturb the representation more than an ordinary topic change" test), Qwen3-4B, layers 12/36 and 30/36:

layer	same-entity (congruent vs contradictory)	cross-entity (congruent vs congruent)
12	0.9974	0.9969
30	**0.9277**	**0.9829**

Replicated across 3 relations × 3 seeds (mid-layer gap ~ 0 on all 9 runs, late-layer gap positive on all 9), and on a second, architecturally different model (Phi-3-mini, layers 11/32 and 27/32; same qualitative pattern, 6/6 relation-model combinations agreeing on direction). Relation-level magnitude is *not* stable across models: Qwen's largest effect is `capital_of` (0.053), Phi-3's largest is `written_by` (0.043) — magnitude depends on the model, not solely the relation.

2.3 Rigidity hypothesis — refuted at fact-level granularity

Proposed mechanism: facts held with a "deeper parametric attractor" should decohere more under override. Tested directly: per-fact teacher-forced confidence in the true tail (bare prompt, no document) against per-fact layer-30 decoherence, n=45 facts pooled across all 3 relations. Spearman rho = 0.006 — no relationship. Confidence vs. override-resistance: rho = −0.225, correct direction but below the ~0.29 threshold n=45 needs for significance. The relation-level magnitude pattern in Section 2.2 is real but not explained by per-fact memorization strength.

2.4 Active-lie hypothesis — refuted with real statistical power

Proposed alternative: decoherence tracks the model successfully *committing* to the false answer, not merely processing the contradiction. Tested on Phi-3 (lower compliance rate gives a naturally large stubborn/compliant split: n=73 stubborn, n=62 compliant, vs. only n=5 stubborn on Qwen):

layer	stubborn (resisted override)	compliant (followed override)	baseline (no conflict)
11	0.0036	0.0036	0.0095
27	**0.1018**	**0.0747**	**0.0623**

Stubborn > compliant > baseline at the late layer — the opposite of what an active-lie mechanism predicts. Resisting an explicit contextual instruction to preserve a parametric prior costs more geometric disruption than complying with it. At the mid layer, both groups score *below* the cross-entity baseline, which is a lexical artifact (swapping the whole subject perturbs shallow-layer tokens more than swapping only the claimed answer for the same subject) rather than evidence about conflict processing.

2.5 Reconciling the mid-layer null with Zhao et al. (2024)

Zhao et al. (arXiv:2410.16090) report ~90% accuracy from a *trained* linear probe detecting conflict-presence at intermediate layers of Llama3-8B — in apparent tension with our untrained mid-layer null (Section 2.2). Replicating their actual method (trained logistic regression, congruent=0/contradictory=1, held out on unseen entities) on our 45-fact Qwen set: AUROC 0.642 at layer 12, 1.000 at layer 30. A trained probe *does* recover a real signal our untrained geometric check missed — the discrepancy is substantially a method difference — but 0.642 falls well short of their ~90%, plausibly due to this replication's much smaller dataset and a single train/held-out split with no cross-validation, not a clean quantitative match.

3. What is and isn't shown

- A real, cross-model, cross-relation late-layer geometric signal under in-context contradiction exists and is not explained by per-fact memorization strength or by whether the model ultimately complies.
- The magnitude of the effect is model- and relation-dependent by a factor of 2–3×; no mechanism tested so far explains that variation.
- The practical framing that survives is **not** "lie detector" — it is closer to a **resistance/friction detector**: it appears to fire more when a model fights its own parametric prior than when it quietly updates to match new context.
- Not shown: what *does* explain the magnitude variation (relation type? entity type? something else); whether this transfers past 3 relation templates and 2 model families; whether it survives natural (non-templated) documents instead of single-sentence synthetic contradictions; a deployed detection threshold (none is proposed here).

4. Honest scope & limitations

- **Small n throughout.** 45 facts, 3 relations, 2–3 seeds per relation. Real effects, not yet large-sample claims.
- **Two working models, two failed premises.** GPT-2-base and TinyLlama-1.1B-Chat do not reliably follow in-context overrides at all and were excluded rather than forced; this itself is worth noting as a capability floor for any downstream use of this signal.
- **Llama-3 untested.** Gated, no license acceptance available in this environment.
- **Synthetic, single-sentence contradictions only.** Real RAG documents are longer, noisier, and multi-fact; this has not been tested on anything but templated single-fact conflicts.
- **No deployed detector.** This note establishes a measurable phenomenon, not a calibrated, cross-domain threshold ready to gate real generations.
- **Prior art.** Zhao et al. (2024) already show conflict-presence is probeable at intermediate layers; this note's distinct contribution is the late-layer resistance-vs-compliance split (Section 2.4) and the negative result on per-fact rigidity (Section 2.3), not the base claim that conflict leaves a residual-stream trace.

5. Reproduce

```
git clone https://github.com/HemanthKiranPolu/nous-decoherence
cd nous-decoherence
pip install -r requirements.txt
python -m nous.verifier_real_llm_test              # 2.1, GPT-2 parametric recall probe
```

```
python -m nous.verifier_in_context_conflict          # 2.2, Qwen single-relation decoherence
python -m nous.verifier_multi_relation_robustness    # 2.2, 3-relation x 3-seed Qwen sweep
python -m nous.rigidity_hypothesis_test              # 2.3, per-fact confidence vs decoherence
python -m nous.verifier_cross_model_phi3             # 2.2, Phi-3 cross-model replication
python -m nous.verifier_compliance_split_phi3        # 2.4, stubborn vs compliant split
python -m nous.replicate_zhao_conflict_probe         # 2.5, trained-probe reconciliation
```

References

- Zhao et al., *Analysing the Residual Stream of Language Models Under Knowledge Conflicts*, arXiv:2410.16090 (2024).
- Li et al., *SHIFT: Gate-Modulated Activation Steering for Knowledge Conflict Mitigation in Retrieval-Augmented Generation*, arXiv:2606.27786 (2026).